

102010694 | Seeed Studio XIAO nRF52840 Sense Plus

SKU: 102010694

Product Overview

Seeed Studio XIAO nRF52840 Sense Plus is a tiny Bluetooth sensor development board powered by the Nordic chipset with an FPU. It includes all capabilities of the standard version such as onboard IMU, PDM microphone, Bluetooth Low Energy 5.4, NFC, and low power consumption. This variant features redesigned castellations and pad layout on the back, providing 9 additional GPIOs for advanced projects requiring expanded I/O via SMD board-to-board soldering. It supports native TensorFlow Lite, making it suitable for embedded Machine Learning and AIoT projects.

Features

High-Performance MCU: Nordic Semiconductor's nRF52840 chip with FPU at 64 MHz, multiple development ports, compatible with Arduino/CircuitPython/MicroPython

Onboard Sensing: Integrated 6-axis IMU and PDM microphone enable motion sensing and audio input, ideal for wearable devices and compact ML applications

Advanced Wireless Connectivity: Bluetooth Low Energy 5.4 with built-in antenna and NFC support

Efficient Power Management: 5 uA deep sleep consumption with integrated Li-ion battery charging

Expanded 20 GPIOs: 9 extra 1.27mm pitch SMD castellation pins beyond standard XIAO nRF52840

Universal Compatibility: Standard backside layout (with aligned battery pads, and debugger pads) for more to-be-released XIAO expansion boards

Classic XIAO Form Factor: 21 x 17.8mm design with 2.54mm pin headers, single-sided mounting for wearables and space-constrained projects

XIAO Accessory Add-ons Compatibility: Natively compatible with all the XIAO accessory add-ons of expansion, connectivity modules (LoRa, CANBus, RS-485), sensors (2-Mic Array, Vision AI, mmWave Motion, Environmental sensing), and actuators (ePaper display, LED Matrix, relay and more).

Production-Ready Design with Fusion PCBA Support As a new addition to the Seeed Studio XIAO family, the XIAO nRF52840 Plus remains the classic XIAO design (thumb-size, SMD, stamp holes). All these features make it a perfect fit for either space-limited applications, or a production-ready System on Module (SoM) for your PCBA designs. Through our Fusion PCBA Sponsorship program, you can not only enjoy the lowest unit prices for all XIAO modules from our Fusion Open Parts Library, but also fabricate and scale your XIAO-based PCBA designs at low cost (30% off) with FREE Design for Manufacturing consulting services, streamlining your journey from prototype to production.

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Product Image

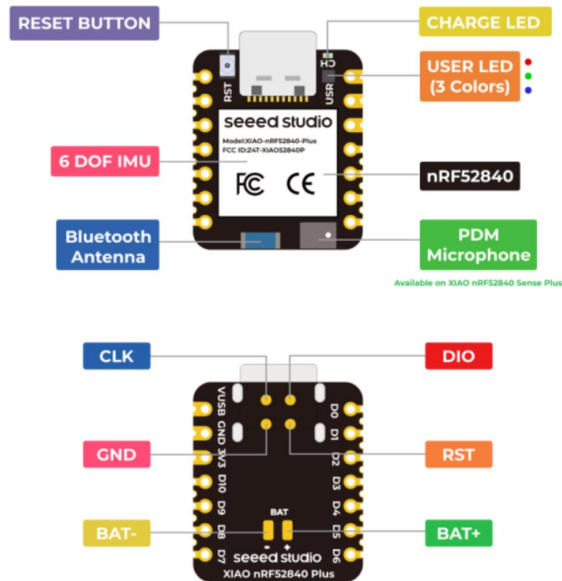


Product image

Specifications

Item	Seeed Studio XIAO nRF52840	Seeed Studio XIAO nRF52840 Sense Plus	Seeed Studio XIAO nRF52840 Plus
Processor	Nordic nRF52840, ARM® Cortex®-M4 32-bit processor with FPU, 64 MHz		
Wireless Connectivity	Bluetooth Low Energy 5.4/Bluetooth Mesh/NFC		
Memory	256KB RAM,1MB Flash 2MB onboard Flash		
Built-in Sensors	N/A	1x6 DOF IMU (LSM6DS3TR-C)1xPDM Microphone	N/A
Interfaces	1xI2C 1xUART1xSPI	1xI2C2xUART2xSPI1xI2S	
PWM/Analog Pins	11/6	20/6	
Onboard Buttons	1x Reset Button1x User Button		
Onboard LEDs	1x 3-in-one LED1x Charge LED		
Battery Charge Chip	BQ25101		
Programming Languages	Arduino/MicroPython/ CircuitPython		

Hardware Overview



Hardware overview image

